cards do not need this.

Enable 'Video RAM Cacheable', only if you have older cards, Disable it for newer cards with more than 16 MB RAM.

Set '8 & 16 bit I/O Recovery Time' to 0 or N/A, if you do not use ISA cards.

Enable 'SDRAM Precharge Control' unless it causes problems with your system.

Set the CAS Latency Time to 2 for fastest performance. This sets the time delay before the SDRAM starts to carry out a read command.

Set the RAS-to-CAS Delay to 2T. This inserts a delay between the RAS (Row Address Strobe) and CAS (Column Address Strobe) signals.

Set the RAS Pre-charge Time to 2T. This is used to set the number of cycles for the RAS to store its charge before the memory refreshes.

Disable 'Memory Hole'.

Disable 'Passive Release'.

Disable 'Delayed Transaction'.

Choose PCI or AGP for 'Init Display First', as per your display card.

Set the KBC Input Clock to 16 MHz. This setting is related to the Super I/O chip.

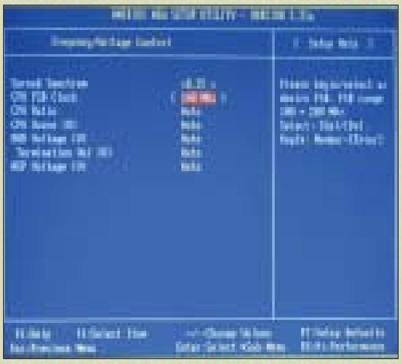
The AGP Aperture size determines how much of your system RAM is shared with your graphics card, to help it function smoothly. Set it according to the amount of RAM on your system: If you have 128 MB or more, set it to 64 MB or 32 MB. If you have less than 128 MB of RAM, set this to half or quarter the amount of RAM you have—if you have 96 MB, set this to 48 MB, or 24 MB, etc. Make sure you allot at least 16 MB to the graphics card.

Overclocking

Caution: Overclocking your CPU will void your warranty, and if done improperly, will shorten the life of your CPU, or even cause a burnout. If you still want to overclock, do so with extreme caution—do not push your CPU beyond a reasonable limit. Done properly, it can yield significant performance boosts.

If you have an Intel motherboard, chances are you cannot overclock your CPU, as Intel desktop motherboards are generally locked. On the other hand, the performance motherboards from Gigabyte and MSI let you tweak a little.

Before overclocking your CPU, you have to change the FSB. Increase the default value from the BIOS in small increments, say from 133 to 135 or 140. Now, change the CPU ratio setting marginally. As all Intel CPUs are locked, this is only applicable to Athlon CPUs. Remember not to change the voltage settings, as it may cause overheating.



Change FSB first, before changing the CDP ratio

Fan please

Ideally, ensure that there are at least two fans within the cabinet. A fan in the front will suck fresh air into the case, while a fan behind will push out the warmer air. An SMPS with two fans will also help.

Cables, cables

Cables from the SMPS are quite often slack; wind up the excess length; if possible, tie up the power cables with string, anchoring them to the top of the cabinet.

Bunch data cables together using duct tape—or split flexible PVC tubing, and place the cables within it. Not only does this keep your cables tangle free, it also keeps them from collecting dust, as well as keeping your cabinet airy. Also, make sure that there are no cables directly above the CPU or graphics card fans.

ATi TWEAKING

OpenGL settings

For optimal performance with the OpenGL API, flip across to the OpenGL tab. Under Main Settings, enable the Custom Settings option.

Under Custom settings, move the slider for the Anti-Aliasing settings to the extreme left, which in this case, is 2X. Select the Performance option here, and set the Anisotropic Filtering slider to the least possible value. Similarly, set the Texture Preference, as well as the Mipmap Detail Level, to High Performance. You can choose to leave the Application Preference options for all the settings disabled. Finally, turn off the Vertical Sync option.